Auteur: Odia Kibor
Studierichting: Informatica
Oefeningenreeks 4A nr. 42.8

$$\int \frac{dx}{x^3 - 7x + 6}$$

nulpunten noemer: 1, 2 en -3

$$\frac{1}{x^3 - 7x + 6} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x+3}$$

$$A = \left.\frac{1}{(x-2)(x+3)}\right|_{x=1} = -\frac{1}{4}$$

$$B = \left.\frac{1}{(x-1)(x+3)}\right|_{x=2} = \frac{1}{5}$$

$$C = \left.\frac{1}{(x-1)(x-2)}\right|_{x=-3} = \frac{1}{20}$$

$$\Rightarrow \int \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x+3} dx = \int -\frac{1}{4(x-1)} + \frac{1}{5(x-2)} + \frac{1}{20(x+3)} dx$$

$$= -\frac{\ln|x-1|}{4} + \frac{\ln|x-2|}{5} + \frac{\ln|x+3|}{20} + C$$

References